



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

Batch: 26th

Date: 12/03/2025

Final Examination

Semester: Spring 2024

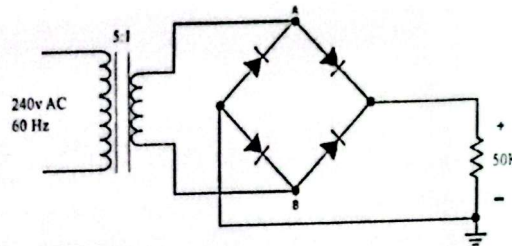
Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

$5 \times 8 = 40$

1.



8

- Calculate the Peak Inverse Voltage (PIV) for a full-wave bridge rectifier.
- Determine the output frequency and provide a logical explanation with a diagram.
- Calculate the rectification efficiency?
- Find the output voltage and current across the resistor 50K.

CLO1 1

CLO1 1

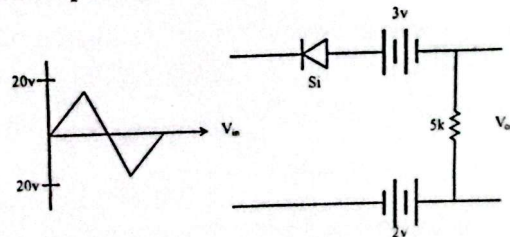
CLO1 2

CLO1 4

2.

- Determine the output signal for the following circuit:

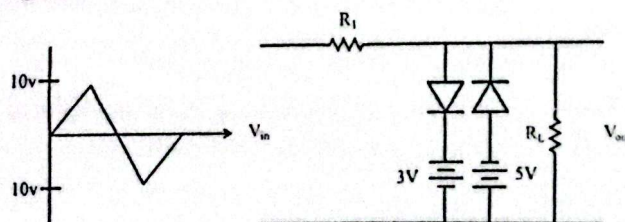
CLO2 4



7

- Draw the output waveform and transfer characteristics of the following circuit:

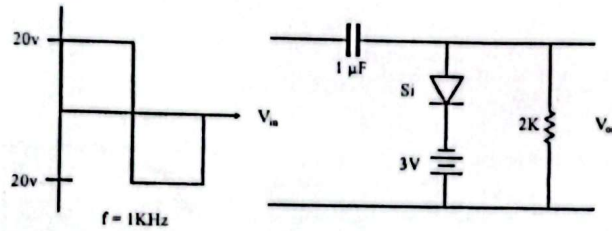
CLO2 4



2. (a) How does the value of the capacitor affect the performance of the clamper circuit? CLO2 3

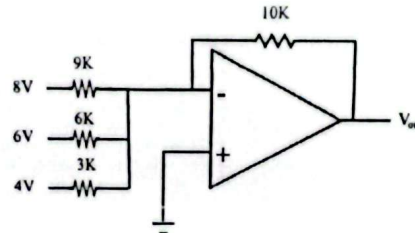
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- (b) CLO2 5



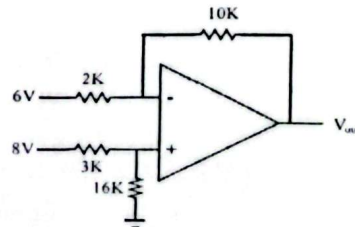
- Determine the Output of the above circuit.
- Analyze whether the capacitance is sufficient to produce the correct output.

4. (a) How can we control the gain of an op-amp? Explain with the concept of feedback circuit. CLO3 2
- (b) Make a comparative analysis between differentiator and integrator op-amp circuits. CLO3 2
- (c) Determine the output for the following circuit: CLO3 4



5. (a) Explain the concept of "virtual ground" in an ideal op-amp. CLO3 2
- (b) What is the Common-Mode Rejection Ratio (CMRR) in op-amps? CLO3 2
- (c) Determine the output voltage of the following circuit: CLO3 4

6



6. (a) Compare inverting and non-inverting op-amp circuits. CLO3 2
- (b) Design a circuit that works as a differentiator for an input signal. CLO3 3
- (c) Design a circuit using op-amps to detect voltages between 3V and 5V. CLO3 3

7

7. (a) Differentiate between BJTs and FETs in terms of their construction, operation, and applications. CLO1 2
- (b) What is the pinch-off condition in a JFET? Explain with diagram. CLO1 3
- (c) Explain the working principle of a MOSFET. CLO1 3

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Final Examination (Fall'24)

PHY 111 Electricity & Magnetism/ PHY 121 Physics II/

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 2 hours

Marks: 40

Answer any 4 (Four) of the following sets of questions

1. (a) What is Hall effect? Show how Hall effect can be used to determine the nature of the charge carriers by drawing the proper figure. 2 CLO1
- (b) A current carrying conductor of a given length is placed in a magnetic field. Derive an expression for the force exerted on the wire by drawing the proper figure. 4 CLO2
- (c) Find the force on each segment of the wire shown in the figure-1(c), if $B = 0.15\text{T}$. Assume that the current in the wire is 5A . 4 CLO3

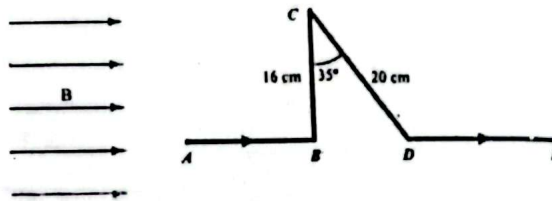


Figure-1(c)

2. (a) State the following terms: 2 CLO1
- I. Biot-Savart law.
- II. Ampere's law.
- (b) Derive an expression for the force between two parallel current carrying conductors. Under what condition the force becomes attractive or repulsive? 4 CLO2
- (c) Two long and fixed parallel wires, A and B are 10 cm apart in air and carry currents of 40 and 20 A, respectively in opposite directions. Determine the resultant flux density (a) on a line midway between the wires and parallel to them, and (b) on a line 8 cm from Wire A and 18 cm from wire B. 4 CLO3
3. (a) What is motional electromotive force? 2 CLO1
- (b) Show that the rate at which heat appears across a closed rectangular loop of width l when it is pulled with respect to a uniform magnetic field of induction B with a constant velocity v is given by $\frac{(B^2 l^2 v^2)}{R}$ where R is the resistance of the loop. 4 CLO2

- (c) A small search coil with an area of 125 mm^2 has 50 turns of very fine wire. This coil is placed between the pole pieces of a small magnet and then suddenly jerked out. If the average induced emf is 0.07 V when the coil is pulled to a field-free region in 60 ms , what is the magnetic intensity between the poles? What was the original flux through each turn? 4 CLO3

4. (a) Define the following terms: 2 CLO1

I. Self-Inductance

II. Mutual Inductance

- (b) State and explain Faraday's law of electromagnetic induction and Lenz's law for the direction of induced emf. 4 CLO2
- (c) A coil of 500 turns has an inductance of 15 mH . Determine (i) the flux produced in the core when a current of 2 A flows in the coil and (ii) the value of the self-induced emf in the coil when the current is changed from $+2 \text{ A}$ to -2 A in 10 ms . 4 CLO3

5. (a) State and explain the basic postulates of Einstein's special theory of relativity. 2 CLO1

- (b) Derive the Lorentz space time transformation formulae. Show that for values of $v \ll c$, Lorentz transformation reduces to the Galilean one. 4 CLO2

- 8 (c) The length of a spaceship is measured to be exactly half of its actual length. Calculate (i) the speed of the spaceship and (ii) the time dilation corresponding to one second on the spaceship. 4 CLO3

6. (a) What is the photo-electric effect? 2 CLO1

- (b) Explain the photo-electric effect using proper figure and also draw the related graph you have acknowledged from the characteristics of photo-electric emission. 4 CLO2

- 8 (c) The photo-electric threshold of copper is $3200 \text{ \AA}$. If ultra-violet light of wavelength $2500 \text{ \AA}$ falls on it, find (a) the maximum kinetic energy of the photo-electrons ejected, (b) maximum velocity of the photo-electrons and (c) the value of the work function. 4 CLO3



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 121: Structured Programming Language

Batch: 26th

Semester: Fall 2024

Date: 19/03/2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

- | | | | | |
|----|-----|---|-------------|-------|
| 1. | (a) | Define Array. Explain how arrays have simplified the declaration process and manipulation of data. | CLO3 | 4 |
| | (b) | Write a C program that prints the sum of the elements of a one-dimensional array. | CLO4 | 4 |
| 2. | (a) | Define loop. Differentiate between counter-controlled and sentinel-controlled loops. | CLO3 | 4 |
| | (b) | Write a C program that takes an integer input from the user and reverses the digits of the number using a while loop. The program should then display the reversed number as output. (e.g. 1234 → 4321) | CLO4 | 4 |
| 3. | (a) | Write two C functions to demonstrate the followings:
i. A function with no arguments and no return value
ii. A function with arguments and one return value | CLO4 | 3 |
| | (b) | What is recursive function? Write a C program that calculates the factorial of a given positive integer using recursion. The program should take an integer as input and return its factorial. | CLO3 & CLO4 | 5 |
| 4. | (a) | Explain the differences between call by value and call by reference with an example. | CLO3 | 2 |
| | (b) | Find out the output of the following programs -
i. <pre>#include<stdio.h> int main() { int a = 5; int *ptr ; ptr = &a; *ptr = *ptr * 3; printf("%d", a); return 0; }</pre> | CLO5 | 3×2=6 |



ii. `#include<stdio.h>`

`int main()`

`{`

`int x = 20, *y, *z;`

`// Assume address of x is 1001 and`

`// integer is 4 byte size`

`y = &x;`

`z = y;`

`*y += 1;`

`*z += 1;`

`x++;`

`printf("x = %d, y = %d, z = %d \n", x, y, z);`

`return 0;`

`}`

iii. `#include<stdio.h>`

`int main()`

`{`

`int x = 10;`

`int *y, **z;`

`y = &x;`

`z = &y;`

`printf("x = %d, y = %d, z = %d\n", x, *y, **z);`

`return 0;`

`}`

5.

(a) Define structure and union in C with their syntaxes.

CLO4

3

(b) Write a C program that defines a structure named Student to store a student's details, including name, roll number, and marks. The program should take input for these details from the user using a loop and then display them.

CLO4

5

6. (a) Convert the following Loop Structures in C -

CLO5

2×2=4

i. The following do-while loop prints the first 5 even numbers. Convert it into a for loop.

`#include <stdio.h>`

`int main() {`

`int i = 2, count = 0;`

`while(count < 5) {`

`printf("%d ", i);`

`i += 2;`

`count++;`

`}`

`return 0;`

- ii. The following C program calculates the sum of the first N natural numbers using a for loop. Convert it into a do-while loop.

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i;
    }
    return 0;
}
```

- (b) Briefly describe the elements of a user-defined function in C with an example. CLO4 4
7. (a) How can you compare two strings in C? Explain with an example. CLO4 3
- (b) Write a C program to print the following number pattern based on user input for the number of rows. For example, when row = 4, the output will be as follows: CLO4 5

```
1
1 2
1 2 3
1 2 3 4
```



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology
Semester Final Examination 2024

Course Code: MAT125
Final Examination
Date: 3/22/2025

Total Time: 2:00 Hours

Course Title: Linear Algebra and Complex Variable
Semester: Fall2024
Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.	Questions	CLOs	Marks
10 Q1.	(a) What is complex number?	CLO1	[1]
	(b) If z_1, z_2 are complex numbers, then show that $ z_1 + z_2 \leq z_1 + z_2 $.	CLO2	[3]
	(c) State De Moivre's theorem and find all values of the following: (i) $x^7 + x^4 + x^3 + 1 = 0$. (ii) $(x + 1)^5 + (x - 1)^5 = 0$.	CLO3	[6]
9 Q2.	(a) What analytic function?	CLO1	[1]
	(b) State and prove the necessary conditions for a function $f(z)$ to be analytic in a region.	CLO2	[5]
	(c) Show that the function $f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}; & \text{when } z \neq 0 \\ 0; & \text{when } z = 0 \end{cases}$, are not analytic at the origin, although Cauchy-Riemann equations are satisfied.	CLO3	[4]
10 Q3.	(a) What is harmonic and harmonic conjugate function?	CLO1	[2]
	(b) Prove that the function $f(z) = e^{z^2}$ satisfied Cauchy-Riemann equations.	CLO2	[4]
	(c) Show that the function $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function. Also find harmonic conjugate of u .	CLO3	[4]
Q4.	(a) Define complex integration.	CLO1	[2]
6	(b) State and prove Cauchy's integral formula.	CLO2	[4]
	(c) Evaluate $\oint_c \frac{e^{2z}}{(z+1)^4} dz$, where c is the circle $ z = 3$.	CLO3	[4]
Q5.	(a) Define singular point, isolated singularities and poles.	CLO1	[2]
	(b) Locate and name all the singularities of $f(z) = \frac{z^8 + z^4 + 5}{(z-1)^3(3z+2)^2}$.	CLO3	[4]
	(c) Calculate $\int_2^{4-2i} (z^2 - 4iz + 6) dz$.	CLO3	[4]
Q6.	(a) Define limit and continuity.	CLO1	[2]
	(b) Find the limit and continuity of $f(z) = \begin{cases} z^2; & \text{when } z \neq i \\ 0; & \text{when } z = i \end{cases}$ at $z = i$.	CLO3	[4]
	(c) Expand $f(z) = \ln(1 + z)$ using Taylor series expansion at $z = 0$.	CLO3	[4]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science & Engineering

BSc in Computer Science & Engineering

ENG 127: English Skills

Final Examination

Fall 2024

26th Batch
24th March 2025

2nd Semester
Time: 2 Hours

Total Marks: 40

Questions:

1. Paraphrase the following passage without changing the original meaning:

An elephant became stuck in the mud at Chunati Wildlife Sanctuary in Banshkhali upazila and was rescued after 22 hours. The operation, which began around 5:00 pm yesterday, was carried out by the Forest Department with assistance from 35 villagers. According to Anisuzzaman Sheikh, a ranger of the Jaldee Range, they were notified of the trapped elephant around 7:00 pm on Wednesday. The female elephant is about 40 years old. After the rescue, the elephant was given saline water to rehydrate.

2. Write the most suitable topic sentence for the following topics:

- a) The Benefits of Reading Books
- b) Your First Day at University
- c) The Impact of Social Media on Society
- d) Role of Exercise in Maintaining Good Health
- e) Environment Pollution

three
three
three

3. Write an essay on *The Role of Technology in Modern Education*.

4. Answer any two of the following questions:

- a) Define Topic Sentence. What are the major characteristics of topic sentence?
- b) What are the differences between a narrative and a descriptive essay? Provide Examples.
- c) What is a book review. Write a format for a book review.

5. Write a review of a book you have recently read.

CLOs	Level	Marks
CLO 2	Understand	5x1=5

CLO 3	Understand	5x1=5
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CLO 4	Apply	12
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CLO 5	Remember	4x2=8
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CLO 4	Apply	10
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